

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0759

Acec Arrowhead 138 kV
October 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0759. The Interconnection Customer proposes a 49.9 MW generating facility interconnecting at the Acec Arrowhead 138 kV substation in the Kentucky area. The request was received on March 13, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 290 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,103,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Central Campus Unm – Henry Creek 138kV
Contingency: Albion Rd – Platte City 138kV
Loading: 157.5% (Rating: 177 MVA)
DFAX: 15.5%
Pre-project loading: 153.1%
2. Facility: Ubc South Campus – Henry Crk 138kV
Contingency: Albion Rd – Platte City 138kV
Loading: 148.5% (Rating: 182 MVA)
DFAX: 15.5%
Pre-project loading: 144.3%
3. Facility: Albion Road – Central Campus Unm 138kV
Contingency: Platte City – Albion Road 138kV line
Loading: 147.8% (Rating: 195 MVA)
DFAX: 15.5%
Pre-project loading: 143.8%
4. Facility: Ayers Island – Kaweah 1 138kV
Contingency: N Appleton – Daisville 138kV
Loading: 133.4% (Rating: 293 MVA)
DFAX: 21.0%
Pre-project loading: 129.8%
5. Facility: Kaweah 1 – Wellington 138kV
Contingency: N Appleton – Davisville 138kV
Loading: 132.5% (Rating: 298 MVA)
DFAX: 21.0%
Pre-project loading: 129.0%

6. Facility: Buckman – Acec Arrowhead 138kV
Contingency: Cedar Tap – Idaho Falls Power 138kV
Loading: 111.7% (Rating: 230 MVA)
DFAX: 49.9%
Pre-project loading: 100.9%
7. Facility: Sidney Coal – Meeker Substation 138kV
Contingency: Spring Crk Rd – Harris 138kV
Loading: 109.5% (Rating: 231 MVA)
DFAX: 7.7%
Pre-project loading: 107.9%
8. Facility: Meeker – Turkey Creek 138kV line
Contingency: Harris – Spring Crk Rd 138kV
Loading: 109.4% (Rating: 219 MVA)
DFAX: 7.7%
Pre-project loading: 107.6%
9. Facility: Cedar Tap – Buckman 138kV
Contingency: Cedar Tap Substation – Idaho Falls Power 138kV
Loading: 108.3% (Rating: 221 MVA)
DFAX: 49.9%
Pre-project loading: 97.0%
10. Facility: Granite City Works – Tule 138kV
Contingency: Davisville – N Appleton 138kV line
Loading: 106.5% (Rating: 196 MVA)
DFAX: 7.1%
Pre-project loading: 104.7%
11. Facility: Longdon – Barnabie Cable Sta 138kV
Contingency: Moss Bluff – Barnabie Cable Sta 138kV Line
Loading: 105.0% (Rating: 484 MVA)
DFAX: 9.5%
Pre-project loading: 104.0%
12. Facility: Turkey Creek Substation – Tom Watkins 138kV
Contingency: Spring Crk Rd – Harris 138kV Line
Loading: 102.4% (Rating: 228 MVA)
DFAX: 7.7%
Pre-project loading: 100.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Central Campus Unm – Henry Creek 138kV	Rebuild line with higher-capacity conductor	\$12,659,000	3% (\$341,000)
Henry Crk Substation – Ubc South Campus 138kV	Rebuild line with higher-capacity conductor	\$6,706,000	3% (\$181,000)
Albion Road – Central Campus Unm 138kV	Rebuild line with higher-capacity conductor	\$3,297,000	3% (\$89,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ayers Island – Kaweah 1 138kV	Rebuild line with higher-capacity conductor	\$40,176,000	3% (\$1,241,000)
Wellington – Kaweah 1 138kV	Rebuild line with higher-capacity conductor	\$6,129,000	3% (\$189,000)
Buckman – Acec Arrowhead 138kV	Rebuild line with higher-capacity conductor	\$18,126,000	9% (\$1,695,000)
Sidney Coal – Meeker 138kV L	Reconductor line	\$1,768,000	2% (\$34,000)
Meeker – Turkey Creek 138kV line	Reconductor line	\$289,000	2% (\$6,000)
Cedar Tap – Buckman 138kV line	Reconductor line	\$546,000	9% (\$51,000)
Tule – Granite City Works 138kV L	Reconductor line	\$11,728,000	2% (\$190,000)
Barnabie Cable Sta – Longdon 138kV	Reconductor line	\$2,533,000	1% (\$26,000)
Turkey Crk – Tom Watkins Substation 138kV	Reconductor line	\$3,100,000	2% (\$60,000)
	Total allocated to project		\$4,103,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.